8.3.3 Using libraries that extend WebDriver

As we mentioned in section 8.2, there are many libraries for different platforms that build on and extend WebDriver, including these:

- *Thucydides* (http://thucydides.info)—Provides powerful WebDriver support for Java and Groovy tests
- *Watir* (http://watir.com)—A powerful Ruby DSL for WebDriver testing
- *WatiN* (http://watin.org)—Provides support for automated web testing in C#
- *Geb* (http://gebish.org)—A Groovy DSL for WebDriver

Rather than trying to cover each library in detail, I'll highlight two of the more compelling features from some of these libraries that make them so popular among test-automation practitioners.

PAGE OBJECTS

All of these libraries provide support for page objects in various forms, making page objects easier to define and use.

Thucydides takes care of instantiating page objects and managing the WebDriver instance, and it provides a number of useful base methods. A simple Thucydides page object looks similar to the standard WebDriver code we've discussed so far:

```
@DefaultUrl("http://localhost:8080/#/welcome")
public class LoginPage extends PageObject {

private WebElement email;
    private WebElement password;
    @FindBy(css = ".btn[value='Sign in']")
    private WebElement signin;

    public void signinAs(String userEmail,
                         String userPassword) {
        email.sendKeys(userEmail);
        password.sendKeys(userPassword);
        signin.click();
    }
}
```

Thucydides page objects extend the PageObject class.

Open the page here by default.

Web elements are defined and used in the normal way.

Thucydides will also instantiate any page object instances in the test code, which simplifies the test code further:

```
LoginPage loginPage;
HomePage homePage;
...
loginPage.open();
loginPage.signinAs(user.getEmail(), user.getPassword());

String welcomeMessage = homePage.getWelcomeMessage();
assertThat(welcomeMessage).isEqualTo("Welcome Jane");
```

Page objects are instantiated automatically.

Open the page at the URL defined by the @DefaultUrl annotation.

Specify the action under test.

Verify the expected outcomes.

WatiN also provides support for page objects in C#. A page object written using WatiN looks like this: